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Master 2: Artificial Intelligence

Module: Big Data Analytics

TP03
Interactive Data Analysis and Visualization with Microsoft Power BI

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Dataset: Data_Entry_2017

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Chapter 1

Introduction

Data analysis and visualization are important for transforming raw data into meaningful information. When datasets contain thousands of records, examining tables alone can make it difficult to identify patterns, compare categories, and summarize the available information. Business intelligence tools such as Microsoft Power BI provide interactive visualizations and statistical indicators that help users explore data more efficiently.

In the medical domain, chest X-ray datasets contain metadata about image records, patient characteristics, and radiological finding labels. Exploring this metadata can help describe the dataset and understand how its records are distributed across different categories. However, descriptive analysis of metadata should not be confused with making independent medical diagnoses.

The objective of this practical work is to build an interactive Power BI dashboard using the `Data_Entry_2017` dataset. The dashboard combines DAX measures, KPI cards, charts, and slicers to provide an accessible overview of the data.

Chapter 2

Objectives

The main objectives of this practical work are:

- Import a CSV dataset into Microsoft Power BI.
- Inspect the data and its column types using Power Query.
- Create summary measures using Data Analysis Expressions (DAX).
- Display key indicators using Card visuals.
- Analyze finding-label categories and X-ray view positions.
- Visualize patient distribution by recorded gender.
- Add interactive slicers for exploring subsets of the data.
- Prepare the project files and documentation for GitHub submission.

Chapter 3

Dataset Description

The project uses the `Data_Entry_2017` dataset, which contains metadata associated with chest X-ray images. The main columns used in the dashboard are described in Table 3.1.

Table 3.1: Main dataset columns used in the analysis

Column	**Description**
`Image Index`	Image identifier or filename.
`Finding Labels`	Labels associated with an image; multiple labels may be separated by the character `|`.
`Patient ID`	Identifier used to distinguish patients.
`Patient Age`	Recorded patient age.
`Patient Gender`	Recorded patient gender.
`View Position`	X-ray view position, such as PA or AP when present.
`Follow-up #`	Follow-up record number.

A patient may have more than one image record. Therefore, the number of image records is not necessarily equal to the number of distinct patients. In addition, a value in `Finding Labels` may contain a combination of labels, which affects how category frequencies should be interpreted.

Chapter 4

Tools and Methodology

4.1 Tools Used

The following tools and technologies were used:

- **Microsoft Power BI Desktop:** dashboard design and visualization.
- **Power Query:** inspection and preparation of imported data.
- **DAX:** creation of measures for summary statistics.
- **CSV:** dataset import format.
- **GitHub:** project storage and sharing.

4.2 Data Import and Preparation

The dataset was imported into Power BI using the Text/CSV connector. The preview was inspected to verify the column names and review the structure of the data. Power Query was used to inspect column types before applying the changes with *Close & Apply*.

For accurate calculations, numeric columns such as `Patient Age` should use a numeric data type. Missing or invalid values should be checked before interpreting summary statistics.

4.3 DAX Measures

Four measures were created to summarize the dataset. The table name used in the formulas below is `Data_Entry_2017`; if the actual table name in Power BI differs, the formulas must be adjusted accordingly.

4.3.1 Total Image Records

```
Total Images =
COUNTROWS('Data_Entry_2017')
```

This measure counts the rows in the table. It represents the number of image records if each row corresponds to one image.

4.3.2 Distinct Patients

```
Total Patients =
DISTINCTCOUNT('Data_Entry_2017'[Patient ID])
```

This measure counts distinct patient identifiers, rather than counting each image record as a separate patient.

4.3.3 Average Patient Age

```
Average Age =
AVERAGE('Data_Entry_2017'[Patient Age])
```

This measure calculates the average of the numeric age values. Blank values are not included in the average; invalid or implausible age values should be reviewed during data preparation.

4.3.4 Male Patients

```
Male Patients =
CALCULATE(
    DISTINCTCOUNT('Data_Entry_2017'[Patient ID]),
    'Data_Entry_2017'[Patient Gender] = "M"
)
```

This measure counts distinct patient identifiers with at least one record whose gender is recorded as `M`. The interpretation assumes that patient gender is sufficiently consistent across records.

Chapter 5

Dashboard Visualizations

5.1 KPI Cards

Card visuals were used to display four summary indicators:

- Total image records.
- Total distinct patients.
- Male patients.
- Average recorded patient age.

In the current dashboard screenshot, the displayed values include approximately 80K image records, 21,606 distinct patients, 12K male patients, and an average age of 55.50 years. The values 80K and 12K are abbreviated displays, so their exact values should be checked in Power BI before being reported as precise counts.

5.2 Finding Labels Bar Chart

A clustered bar chart was created with `Finding Labels` as the category and the count of `Image Index` as the value. The chart is titled *Top 10 Most Common Findings*.

The chart compares the frequencies of the recorded finding-label categories. If a record contains a combined value such as `Label A|Label B`, Power BI treats that complete combination as one category unless the labels are separated during data preparation. Therefore, this chart should be interpreted as a distribution of recorded label categories, not necessarily as an independent count of every condition.

5.3 X-ray View Position Chart

A clustered column chart compares the number of image records by `View Position`. The current dashboard displays the categories PA and AP. This chart helps compare the frequency of image records across the available view positions.

5.4 Patient Gender Donut Chart

A donut chart was created using `Patient Gender` as the legend and the distinct count of `Patient ID` as the value. It provides a visual comparison of the distinct patients in the recorded gender categories. The result depends on the completeness and consistency of the gender information.

5.5 Interactive Slicers

The dashboard includes the following slicers:

- **Patient Gender:** filters the data by recorded gender.
- **View Position:** filters the data by X-ray view position.
- **Patient Age:** filters the data by age range.

Slicers allow users to explore subsets of the dataset without manually changing the charts. Their effect depends on the visual interactions configured in Power BI.

5.6 Dashboard Screenshot

The final dashboard screenshot can be added to the project folder as `dashboard.png`. Upload the image to Overleaf and uncomment the figure code below to include it in the report.

Chapter 6

Results and Discussion

The dashboard provides a descriptive overview of the chest X-ray metadata. The KPI cards summarize image records, distinct patients, male patients, and average recorded age. In the current screenshot, the dashboard displays approximately 80,000 image records, 21,606 distinct patients, and an average age of 55.50 years. The male-patient value is displayed in abbreviated form as 12K.

The finding-label bar chart compares frequently occurring recorded categories, while the view-position chart compares the PA and AP categories. The gender donut chart summarizes the distribution of distinct patient identifiers across the recorded gender categories. Interactive slicers allow users to explore how these summaries change when different filters are selected.

These values should be treated as descriptive statistics of the loaded dataset. The abbreviated values should be verified using the full values in Power BI before final submission. Moreover, finding-label combinations and possible data-quality issues should be considered when interpreting the charts.

Chapter 7

Limitations and Possible Improvements

The dashboard has several limitations:

1. A patient may have multiple image records; image counts and patient counts must therefore be distinguished.
2. The `Finding Labels` field may contain multiple labels in a single value, which can cause combined categories to be counted together.
3. Missing, inconsistent, or incorrectly typed data may affect the summary measures.
4. The dashboard describes metadata and does not independently establish medical diagnoses.

Possible improvements include separating combined finding labels into individual labels using Power Query, validating age values, displaying exact KPI values, improving chart labels, and checking that all slicers interact with the intended visuals.

Chapter 8

Conclusion

This practical work demonstrates how Microsoft Power BI can transform structured metadata into an interactive dashboard. The project includes data import, data inspection, DAX measures, KPI cards, charts, and slicers. Together, these elements provide a convenient overview of image records, patient characteristics, finding-label categories, and X-ray view positions.

The work also highlights the importance of choosing appropriate aggregation methods and interpreting visualizations according to the structure and limitations of the dataset. The resulting dashboard supports descriptive data exploration and provides a foundation for further data-cleaning and analysis tasks.